

Table Data Visualization



Module Outline

- Visualization
 - Effective Design Techniques
 - Tables
 - Crosstabulation
 - Pivot Tables

Data Visualization

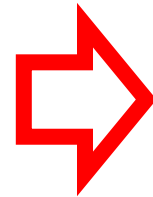
- First step in trying to interpret data is often to visualize the data in some way
- Data visualization can be as simple as creating a summary table for the data, or it could require generating charts to help interpret, analyze and learn from the data.
- Data visualization is also important in conveying your analysis to others.
- It is very helpful for identifying data errors and for reducing the size of your data set by highlighting important relationships and trends in the data
- Person analyzing the data has to make the analysis simple for others to understand. Proper data visualization techniques greatly improve the ability of the decision maker to interpret the analysis easily
- Decades of research studies in psychology and other fields show that human mind can process visual images such as charts faster than rows of numbers.

Effective Design Techniques

- Data-Ink Ratio (DIR), first described by Edward R. Tufte in 2001 in his book “The Visual Display of Quantitative Information.”
- DIR measures the proportion of what the “Tufte” terms “data-ink” to the total amount of ink used in a table or chart.
- Data ink is the ink used in a table or chart that is necessary to convey the meaning of the data to the audience.
- Non data ink is ink used in a table or chart that serves no useful purpose in conveying the data to the reader.

Effective Design Techniques

Daily Sales			
Day	Sales	Day	Sales
1	18	11	33
2	21	12	21
3	25	13	40
4	33	14	26
5	24	15	34
6	19	16	37
7	24	17	23
8	13	18	38
9	33	19	15
10	36	20	14



Daily Sales			
Day	Sales	Day	Sales
1	18	11	33
2	21	12	21
3	25	13	40
4	33	14	26
5	24	15	34
6	19	16	37
7	24	17	23
8	13	18	38
9	33	19	15
10	36	20	14

Tables

- Tables should and are more appropriate to use when
 - The reader needs to refer to specific numerical values
 - The reader needs to make precise comparison between different values and not just relative comparisons
 - The values being displayed have different units or very different magnitudes.

	Month						Total
	1	2	3	4	5	6	
Sales	\$ 148,009.00	\$ 189,976.00	\$ 243,934.00	\$ 311,877.00	\$ 375,879.00	\$ 136,038.00	\$ 1,405,713.00
Cost	\$ 46,622.00	\$ 36,607.00	\$ 15,864.00	\$ 44,555.00	\$ 31,188.00	\$ 47,439.00	\$ 222,275.00

Tables – General Guidelines

- Increase Data Ink Ratio in tables by minimizing grid lines around the table, and a large proportion of the ink used in the table is used to convey the information (the actual numbers)
- As a general rule, one should avoid using vertical lines in a table unless they are necessary for clarity. Horizontal lines are generally necessary only for separating column titles from data values or when indicating that a calculation has taken place.
- In larger tables vertical lines or light shading can be useful to help the reader differentiate the column and rows.

Tables – General Guidelines

	Month						
	1	2	3	4	5	6	Total
Sales	\$ 148,009.00	\$ 189,976.00	\$ 243,934.00	\$ 311,877.00	\$ 375,879.00	\$ 136,038.00	\$ 1,405,713.00
Cost	\$ 46,622.00	\$ 36,607.00	\$ 15,864.00	\$ 44,555.00	\$ 31,188.00	\$ 47,439.00	\$ 222,275.00
Profit	\$ 101,387.00	\$ 153,369.00	\$ 228,070.00	\$ 267,322.00	\$ 344,691.00	\$ 88,599.00	\$ 1,183,438.00

Design A

⊕							
	Month						
	1	2	3	4	5	6	Total
Sales	\$ 148,009.00	\$ 189,976.00	\$ 243,934.00	\$ 311,877.00	\$ 375,879.00	\$ 136,038.00	\$ 1,405,713.00
Cost	\$ 46,622.00	\$ 36,607.00	\$ 15,864.00	\$ 44,555.00	\$ 31,188.00	\$ 47,439.00	\$ 222,275.00
Profit	\$ 101,387.00	\$ 153,369.00	\$ 228,070.00	\$ 267,322.00	\$ 344,691.00	\$ 88,599.00	\$ 1,183,438.00

Design B

Tables – General Guidelines

	Month						Total
	1	2	3	4	5	6	
Sales	\$ 148,009.00	\$ 189,976.00	\$ 243,934.00	\$ 311,877.00	\$ 375,879.00	\$ 136,038.00	\$ 1,405,713.00
Cost	\$ 46,622.00	\$ 36,607.00	\$ 15,864.00	\$ 44,555.00	\$ 31,188.00	\$ 47,439.00	\$ 222,275.00
Profit	\$ 101,387.00	\$ 153,369.00	\$ 228,070.00	\$ 267,322.00	\$ 344,691.00	\$ 88,599.00	\$ 1,183,438.00

Design C

	Month						Total
	1	2	3	4	5	6	
Sales	\$ 148,009.00	\$ 189,976.00	\$ 243,934.00	\$ 311,877.00	\$ 375,879.00	\$ 136,038.00	\$ 1,405,713.00
Cost	\$ 46,622.00	\$ 36,607.00	\$ 15,864.00	\$ 44,555.00	\$ 31,188.00	\$ 47,439.00	\$ 222,275.00
Profit	\$ 101,387.00	\$ 153,369.00	\$ 228,070.00	\$ 267,322.00	\$ 344,691.00	\$ 88,599.00	\$ 1,183,438.00

Design D

Tables – General Guidelines

- Columns of numerical values in a table should be right-aligned. The final digit of each number should be aligned in the column. This makes easy to see differences in the magnitude of values.
- If you are showing digits to the right of the decimal point, all values should include the same number of digits to the right of the decimal point.
- Use only the number of digits that are necessary to convey the meaning in comparing the values, there is no need to include additional digits if they are not meaningful for comparisons.
- It is generally best to left-align text values within a column table. In some cases you may prefer to center the text, but you should do this only if text values are approximately the same length. Otherwise, aligning the first letter of each data entry promotes readability.
- Column headings should either match the alignment of the data in the column or be centered over the values

Cross tabulation

- It is used for describing data of several variables.
- Cross tabulation is usually performed on **categorical data** that can be divided into mutually exclusive groups.
- The most important thing to remember about categorical data is that a categorical data point cannot belong to more than one category.
- Cross tabulations are used to examine relationships within data that may not be readily apparent. Cross tabulation is especially useful for studying market research or survey responses. Cross tabulation of categorical data can be done with through tools such as SPSS, SAS, and Microsoft Excel.

Pivot Tables

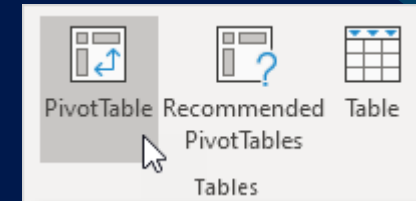
- A crosstabulation in Excel is known as a PivotTable.
- A PivotTable is a powerful tool to calculate, summarize, and analyze data that lets you see comparisons, patterns, and trends in your data.
- Pivot tables are one of Excel's most powerful features. A pivot table allows you to extract the significance from a large, detailed data set.

Create a pivot table

- Data Set

	A	B	C	D	E	F
1	Order ID	Product	Category	Amount	Date	Country
2	1	Carrots	Vegetables	\$4,270	06/01/2016	United States
3	2	Broccoli	Vegetables	\$8,239	07/01/2016	United Kingdom
4	3	Banana	Fruit	\$617	08/01/2016	United States
5	4	Banana	Fruit	\$8,384	10/01/2016	Canada
6	5	Beans	Vegetables	\$2,626	10/01/2016	Germany
7	6	Orange	Fruit	\$3,610	11/01/2016	United States
8	7	Broccoli	Vegetables	\$9,062	11/01/2016	Australia
9	8	Banana	Fruit	\$6,906	16/01/2016	New Zealand
10	9	Apple	Fruit	\$2,417	16/01/2016	France
11	10	Apple	Fruit	\$7,431	16/01/2016	Canada
12	11	Banana	Fruit	\$8,250	16/01/2016	Germany
13	12	Broccoli	Vegetables	\$7,012	18/01/2016	United States
14	13	Carrots	Vegetables	\$1,903	20/01/2016	Germany
15	14	Broccoli	Vegetables	\$2,824	22/01/2016	Canada
16	15	Apple	Fruit	\$6,946	24/01/2016	France
17	16	Banana	Fruit	\$2,320	27/01/2016	United Kingdom
18	17	Banana	Fruit	\$2,116	28/01/2016	United States
19	18	Banana	Fruit	\$1,135	30/01/2016	United Kingdom
20	19	Broccoli	Vegetables	\$3,595	30/01/2016	United Kingdom
21	20	Apple	Fruit	\$1,161	02/02/2016	United States
22	21	Orange	Fruit	\$2,256	04/02/2016	France
23	22	Banana	Fruit	\$1,004	11/02/2016	New Zealand
24	23	Banana	Fruit	\$3,642	14/02/2016	Canada
25	24	Banana	Fruit	\$4,582	17/02/2016	United States
26	25	Beans	Vegetables	\$3,559	17/02/2016	United Kingdom
27	26	Carrots	Vegetables	\$5,154	17/02/2016	Australia
28	27	Mango	Fruit	\$7,388	18/02/2016	France
29	28	Beans	Vegetables	\$7,163	18/02/2016	United States
30	29	Beans	Vegetables	\$5,101	20/02/2016	Germany

1. Click any single cell inside the data set.
2. On the Insert tab, in the Tables group, click PivotTable.



- The following dialog box appears. Excel automatically selects the data for you. The default location for a new pivot table is New Worksheet.
3. Click OK.

A screenshot of the 'Create PivotTable' dialog box in Excel. The dialog has a title bar with a question mark and a close button. It contains several sections:

- Choose the data that you want to analyze**: The 'Select a table or range' radio button is selected. The 'Table/Range' text box contains 'Sheet1!\$A\$1:\$F\$214'. The 'Use an external data source' radio button is unselected, and the 'Choose Connection...' button is disabled. The 'Use this workbook's Data Model' radio button is unselected.
- Choose where you want the PivotTable report to be placed**: The 'New Worksheet' radio button is selected. The 'Existing Worksheet' radio button is unselected, and the 'Location' text box is empty.
- Choose whether you want to analyze multiple tables**: The 'Add this data to the Data Model' checkbox is unselected.

At the bottom right, there are 'OK' and 'Cancel' buttons. A mouse cursor is pointing at the 'OK' button.

Create a pivot table

- Drag fields
- The PivotTable Fields pane appears. To get the total amount exported of each product, drag the following fields to the different areas.
- 1. Product field to the Rows area.
- 2. Amount field to the Values area.
- 3. Country field to the Filters area.
- Below you can find the pivot table. Bananas are our main export product. That's how easy pivot tables can be!

PivotTable Fields [X]

Choose fields to add to report: [Settings]

Search [Search Icon]

☐ Order ID
☒ **Product**
☐ Category
☒ **Amount**
☐ Date
☒ **Country**

Drag fields between areas below:

Filters	Columns
Country [Dropdown]	

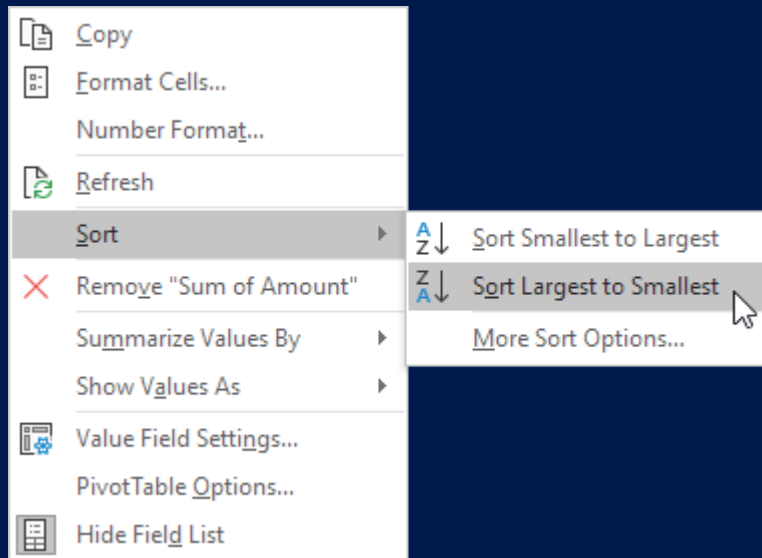
Rows	Values
Product [Dropdown]	Sum of Amou... [Dropdown]

☐ Defer Layout Update [Update]

	A	B	C
1	Country	(All)	
2			
3	Row Labels	Sum of Amount	
4	Apple	191257	
5	Banana	340295	
6	Beans	57281	
7	Broccoli	142439	
8	Carrots	136945	
9	Mango	57079	
10	Orange	104438	
11	Grand Total	1029734	
12			

Sort Pivot Table

- To get Banana at the top of the list, sort the pivot table.
- 1. Click any cell inside the Sum of Amount column.
- 2. Right click and click on Sort, Sort Largest to Smallest.



	A	B	C
1	Country	(All)	
2			
3	Row Labels	Sum of Amount	
4	Banana	340295	
5	Apple	191257	
6	Broccoli	142439	
7	Carrots	136945	
8	Orange	104438	
9	Beans	57281	
10	Mango	57079	
11	Grand Total	1029734	
12			

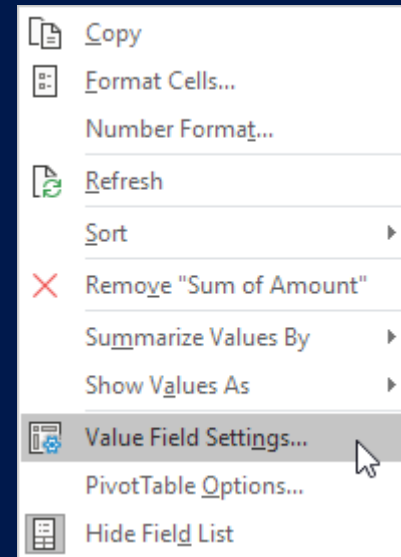
Filter Pivot Table

- Because we added the Country field to the Filters area, we can filter this pivot table by Country. For example, which products do we export the most to France?
- 1. Click the filter drop-down and select France.
- Result. Apples are our main export product to France.
- Note: you can use the standard filter (triangle next to Row Labels) to only show the amounts of specific products.

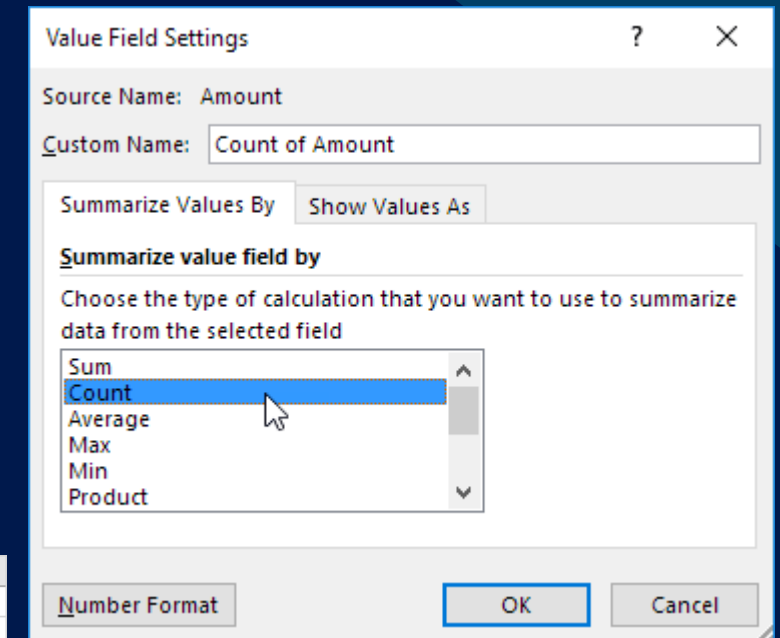
	A	B	C
1	Country	France	
2			
3	Row Labels	Sum of Amount	
4	Apple	80193	
5	Banana	36094	
6	Carrots	9104	
7	Mango	7388	
8	Broccoli	5341	
9	Orange	2256	
10	Beans	680	
11	Grand Total	141056	
12			

Pivot Table Change Summary Calculation

- By default, Excel summarizes your data by either summing or counting the items. To change the type of calculation that you want to use, execute the following steps.
- 1. Click any cell inside the Sum of Amount column.
- 2. Right click and click on Value Field Settings.
- 3. Choose the type of calculation you want to use. For example, click Count.
- 4. Click OK.



	A	B	C
1	Country	France	
2			
3	Row Labels	Count of Amount	
4	Apple	16	
5	Banana	7	
6	Carrots	1	
7	Mango	1	
8	Orange	1	
9	Beans	1	
10	Broccoli	1	
11	Grand Total	28	
12			



Reference

- Course Notes from Eugene Rex L. Jalao, Ph.D. Business Intelligence Training of University of the Philippines National Engineering Center.
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